PROJECT SUMMARY

Project Background

Big Box Warehouses (BBW) is in the process of modernizing its stores. One critical aspect BBW must improve in its stores is the checkout process. One of the main customers complaints the company receives is the long lines at the checkouts, with customers frequently reporting waiting times in checkout queues of 30 minutes and above during the busier periods of the day.

BBW is testing two new models of checkout stations from the CAU Corporation: one model of cashier checkout station and one model of self-checkout station. Before this project, BBW did not use self-checkout stations on any of its stores. This report summarizes the observations and recommendations resulting from proof-of-concept implementation of the new checkout stations in BBW's store # 011.

Store #011 previously had 12 cashiers checkout lanes. This project replaced 8 of those with 4 self-checkout lanes and 6 new cashier checkout lanes. There is space for two self-checkout stations on each self-checkout lane. CAU Corporation provides the option of setting up a single queue for all the self-checkout stations or one queue for each self-checkout lane, but only the individual queue per lane option was tested.

Project Observations and Results

This section includes the most important observations and results from the project.

- The new cashier and self-checkout station models require less space than the current cashier checkout stations. Up to 25% more checkout lanes (cashier of self) can be fit in the same space in comparison to the current ones.

- A 10% reduction in checkout times were observed in the new cashier checkout stations versus the current ones.
- A limit of 20 items for the self-checkout stations was found to be optimal.
- A higher average checking out time was observed for the self-checkout stations, however the 2 stations per lane allows more customers to check out per lane-hour than on the cashier checkout stations.
- CAU Corporation station designs require just one employee per up to 12 self-checkout stations to help customers, and one employee per cashier checkout station.
- It was observed that customers with 20 items or less utilized cashier checkouts if the cashier was idle, regardless of the availability of the self-checkout stations. Otherwise, they use the lane, cashier or self-checkout, with the shorter queue.
- The project personnel visited stores where the single queue for the self-checkout stations was used. It was observed that customers with eligible number of items went directly to the self-checkout queue if no cashier was idle, regardless the length of the queues.

Final Recommendations

This project team recommends the implementation of CAU corporation checkout stations on all BBW's stores. A discrete simulation study should be conducted for each store to determine the best checkout area configuration according to the store customer traffic. The use of a single queue for all the self-checkout stations should be evaluated in the study. The self-checkout stations should be limited to 20 items per customer.